Basics must improve to reduce the burden of hamstring muscle injuries

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INTRODUCTION

Despite a variety of injury prevention programmes and a rapid increase in research over the past decade, the incidence of hamstring muscle injuries (HSI) in many competitive sports has not decreased.[1] Risk factors are multifactorial and may include anthropometry, game volume and intensity, competitive timetabling (congestion), length of career, international travel and duties and of course a history of prior injury and compliance to adequate rehabilitation. While some risk factors are not modifiable, inappropriate decision-making and/or non-adherence to rehabilitation practices may partially explain the increased incidence (and reoccurrence of HSI).[2] Clinicians managing athletes with HSI require the best available evidence to support their management in optimising return to play. Unfortunately, evidence in key areas remains weak. For example, criteria used to guide return to full sprinting do not encompass the differing roles of each hamstring muscle in sprinting. Furthermore, agreement is lacking on when to consider surgical intervention to restore connective tissue architecture, tension and reduce reinjury risk in severe HSI. Given the increasing incidence and burden of HSI, together with what appears to be suboptimal injury prevention and rehabilitation evidence, the sports and exercise medicine community should be asking 'have we even got the basics nailed down?'

HOW CAN HSI MANAGEMENT IMPROVE IN THE ABSENCE OF EVIDENCE?

In the absence of high-quality robust data, seeking expert opinion is helpful in shaping research hypotheses and identifying commonalities in appropriate decision-making. A Delphi method can determine if current practice is aligned with current evidence and informs research questions if significant gaps between evidence and practice are identified.[3]

The London International Consensus and Delphi study on Hamstring Injuries, after undertaking review,[4] set out to ascertain consensus on what represents optimal HSI management in athletes. Experts identified three key topics where evidence was lacking. These were (1) defining the problem: diagnosis and classification of HSI, (2) the role of restoring anatomy: surgical decision-making and (3) recovery and reducing the risk of recurrence: HSI rehabilitation including return to high-speed running and sport. After 2 days of vigorous debate and voting, we produced a three-part series of consensus statements, which were sent to a larger global body of HSI experts.[5–7] While we present the results in *British Journal of Sports Medicine*, some answers highlight just how far we are from doing the basics well.

DIAGNOSIS AND CLASSIFICATION

Our results suggest clinicians managing HSI prioritise the British Athletic Muscle Injury Classification[8] but use the Munich[9] and Barcelona[10] systems when diagnostic gaps occur. MRI is the imaging modality of choice with few clinicians prioritising ultrasound. Neither modality, however, is optimal for monitoring rehabilitation progression or deciding readiness to return to sport. Experts agreed that muscle injury classifications must evolve to include individual muscles, intramuscular injuries, mechanism of injury, sporting demand, functional criteria and patient-related outcome measures. Finally, and importantly, the validity and reliability of classification systems to prognosticate and guide treatment decision-making must improve.

SURGICAL INDICATION AND MANAGEMENT

Most experts agreed that surgical criteria within HSI classification systems must improve. A minority of hamstring injuries require surgical repair, but certain factors aid management. Key surgical indications include gapping at the zone of the injury. Elite-level athletes with severe grade musculotendinous junction and distal hamstring injuries demanding return to previous sporting level and low risk of reinjury, merit surgical consideration as well as the non-surgical rehabilitation to achieve their specific demands. Emerging evidence supports this consensus with improved hamstring strength and return to sport in cohort studies, but much more research is required.

REHABILITATION

Our experts agreed that differences in hamstring musculotendinous tissue, anatomy and functional roles should direct rehabilitation prescription after hamstring injury. In the early phase of injury, most advocate protection of injured tissue from loading at length and elastic loads (ie, high strain and strain rate loads), but there is a difference of opinion in the order and types of load/contraction when initiating rehabilitation (ie, concentric vs isometric loads). Experts use an integrated assessment of symptoms, strength and response to previous loading as criteria for progressing, dosing exercise and deciding on safety to return to running and sport. Other criteria such as flexibility and special return to sport tests are used less widely. On criteria related to pain, experts suggest some activities should be pain free through rehabilitation (ie, sprinting), but a pain threshold approach can be permitted in certain activities. In later stage rehabilitation, experts aim to achieve full outer length and eccentric strength as key criteria for return to running and sport. Experts focus on capacity and demands required for competition when deciding the end rehabilitation goals and timing of return to sport. Experts monitor and test athletes through rehabilitation and use modalities such as global positioning systems to give sport-specific information on loading/running dosages, speed and RTS readiness.

In all areas of this Delhi consensus, it is clear that high-quality hamstring research in elite populations is limited. Effective extrapolation of data from subelite populations or from low to high hamstring demand sports is suboptimal. The London consensus papers summarise important research questions related to HSI that we must pursue with a strong interdisciplinary and international focus. This will require careful data collection, robust studies where appropriate, hamstring registry data collation and analysis.

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